

SPECTROSCOPY: Spectroscopy may be defined as the interaction between the matter and electromagnetic radiations.

There are two types of spectrums:

Emission spectra: Molecules give emission spectra when subjected to intense heat or electric discharge. The molecules obtain the necessary energy to become excited. On returning to their lower energy state, molecules may emit radiation, which is the result of a transition of a molecule from an excited state to one of lower energy, usually the ground state. This excess energy is emitted as photon and the corresponding frequency is recorded as the emission spectrum.

Absorption spectra: When a substance is irradiated with electromagnetic radiation, the energy of the incident photons may be transferred to the molecules raising them from the ground state to an excited state. This process is known as absorption and the resultant spectrum is known as absorption spectrum. Energy absorption occurs only when the energy difference between the ground state and higher energy level is exactly matched by the energy of the incident electromagnetic radiation.

Representation of absorption spectrum: When electromagnetic radiation interacts with the molecules, then some of wavelengths (or frequency or wavenumbers) of radiation are absorbed while the rest of it is transmitted. Consequently, we can plot either

- 1) Absorbance (A) versus λ or frequency or wave number
- 2) Percent transmittance (%T) versus λ or frequency or wave number.

Principal of IR spectroscopy: The IR spectroscopy theory utilizes the concept that molecules tend to absorb specific frequencies of light that are characteristic of the corresponding structure of the molecules. The energies are reliant on the shape of the molecular surfaces, the associated vibronic coupling, and the mass corresponding to the atoms.

For instance, the molecule can absorb the energy contained in the incident light and the result is a faster rotation or a more pronounced vibration.

INFRARED (IR) SPECTROSCOPY

OR

VIBRATIONAL SPECTROSCOPY

- Infrared spectroscopy is used in identification of functional groups in pure compounds.
- Infra-red (IR) does not have sufficient energy to induce electronic transition as seen in UV spectroscopy. When molecule absorbed electromagnetic radiation in IR region, undergoes vibrational or a rotational transition which causes net change in the dipole moment in the molecule (**IR active**, for example HCl, CO etc), if dipole moment does not change in molecules, then they are **IR inactive** (for example: O₂, H₂, N₂ etc.) means they does not absorb IR radiation. IR region ranges from **4000-400 cm⁻¹**.
- If the frequency of IR radiation matched with the vibrational frequency of molecule, then molecule absorb radiation.
- IR spectroscopy based on **Hooke's law**, suppose two atoms or masses are connected through spring (bond), then frequency of vibration can be represented by following equation:

$$\nu = \frac{1}{2\pi} \sqrt{\frac{\bar{\kappa}}{\mu}} \text{ or } \bar{\nu} = \frac{1}{2\pi c} \sqrt{\frac{\bar{\kappa}}{\mu}}$$

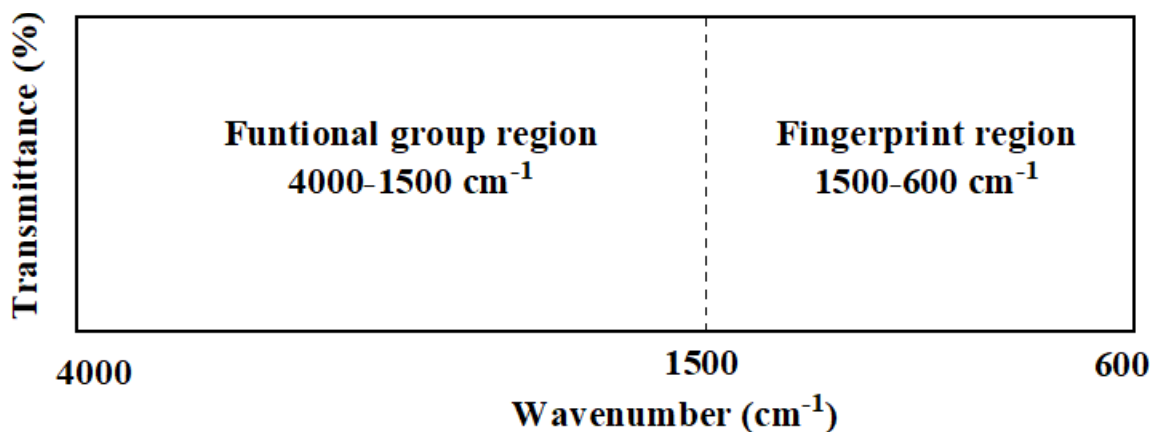
Where, κ is force constant of the bond, $\bar{\nu}$ is wave-number (cm⁻¹), ν is the frequency, c is speed of light and μ is reduce mass (m_1 and m_2 are the masses of atoms)

$$\mu = \frac{m_1 \times m_2}{(m_1 + m_2)}$$

- Stronger the bond, greater the value of force constant (κ), higher the frequency vibration or wave-number (cm⁻¹). For example,

	C – C	C = C	C ≡ C
force constant	5 x 10 ⁵	10 x 10 ⁵	15 x 10 ⁵ ($\frac{\text{dynes}}{\text{cm}}$)
wavenumber (cm ⁻¹)	1200	1650	2100

- IR spectrum is divided mainly into two region as follows;



FUNDAMENTAL VIBRATION

These vibrations are arising when molecule promoted from ground state to lower excited state. The fundamental vibrations for linear and non-linear molecules are determined by following way:

Molecule	Degree of freedom
Linear	$3n-5$
Non-linear	$3n-6$

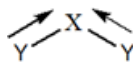
Where, 'n' is number of atom present in the molecule

The vibrations discussed below are fundamental vibrations.

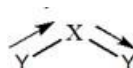
- a) Stretching vibration: Distance between two atoms increase and decrease but bond angle remains constant.

Types of stretching vibrations

- i) Symmetric stretching vibration: In this case both the atoms stretched or compressed in same direction.



- ii) Asymmetric stretching vibration: In this vibration one atom undergoes stretching and other atom undergoes compression and vice versa.

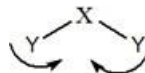


- b) Bending vibrations: Distance between two atoms remains constant but bond angle changes. These vibrations can occur either in plane or out of plane.

Types of bending vibrations

1) In plane bending vibrations:

- i) Scissoring: both the atom move towards each other just like scissor.

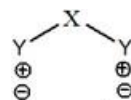


- ii) Rocking: both the atoms move in same direction, either in left side or right side.

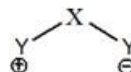


2) Out of plane bending vibrations:

- i) Wagging: both the atom move up and down with respect to central atom.

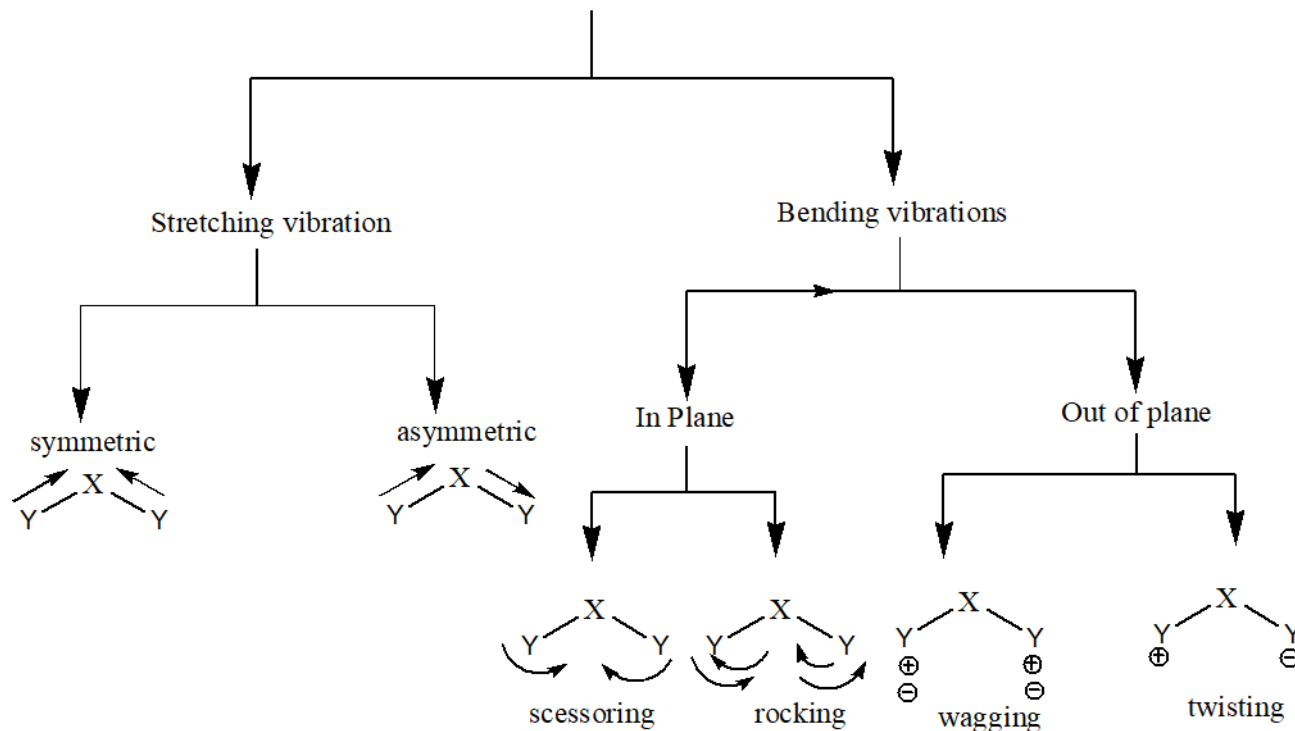


- ii) Twisting: one atom move up and other atom move down with respect to central atom.



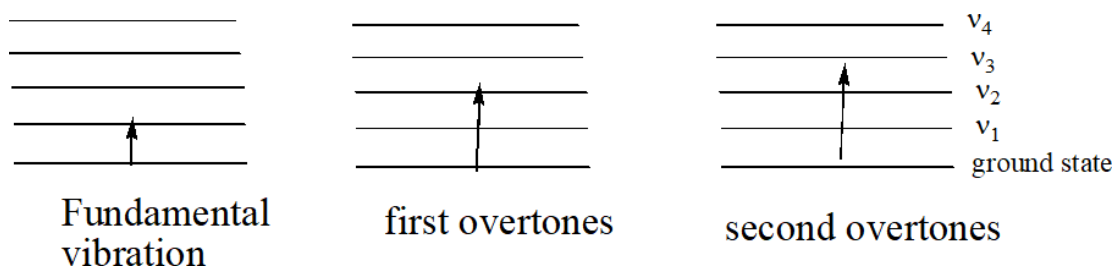
central atom.

Fundamental Vibrations in XY_2 molecule



OVERTONES AND COMBINATION BANDS

When molecule absorbed electromagnetic radiation in IR region, and then molecule promoted from ground state to second, third or even fourth vibrational excited state. These bands are known as Overtones. The intensity of these bands is very weak. It is helpful in characterization of aromatic compounds.



When two fundamental vibrational frequencies ($v_1 + v_2$) in a molecule couple to give rise to a new vibrational frequency within the molecule, it is known as combination band.

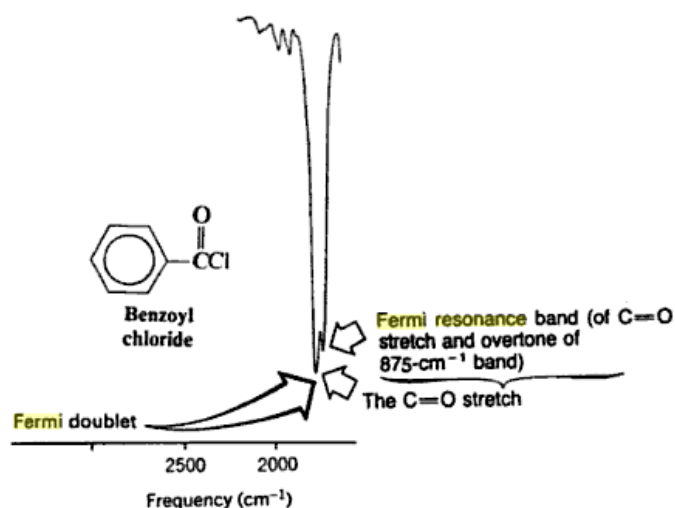
COUPLED VIBRATIONS

The coupled vibrations are observed in group like $-\text{CH}_2$, NH_2 etc. In these groups same atoms are attached to the central atom. When $-\text{CH}_2$ undergoes vibration by the absorption of IR radiation, due to internal perturbation, energy of one C-H bond is transfer to neighboring C-H bond which enhance its vibrational frequency. Therefore, two stretching vibrational frequencies for $-\text{CH}_2$ group is observed at 2950 cm^{-1} (asymmetric stretching) and 2860 cm^{-1} (symmetric stretching).

FERMI RESONANCE

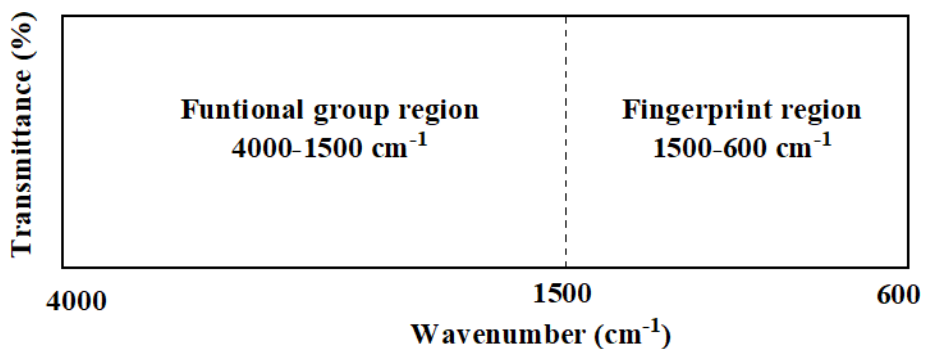
When fundamental vibration coupled with overtones or combination band, the coupled vibration is called Fermi resonance or when molecule absorb IR radiation then it transfers its energy or intensity from fundamental vibration to overtones, then Fermi resonance is observed. As we know that the intensity of overtones band is very weak as compare to fundamental vibrations. But, due to transfer of energy, the strong band is observed for overtones along with the fundamental frequency. Fermi resonance is generally observed in carbonyl groups. For example, in benzoyl chloride $-\text{C}=\text{O}$ stretching vibration observed at 1790 cm^{-1} and 1745 cm^{-1} . The lower frequency band at 1745 cm^{-1} is observed due to combination of

overtone of CH bending vibration at 875 cm^{-1} with the fundamental vibration of C=O stretching.



FINGERPRINT REGION

The region from $1500\text{-}600\text{ cm}^{-1}$ in IR spectrum is known as Fingerprint region. In this region number of bending vibration is more than the number of stretching vibration.



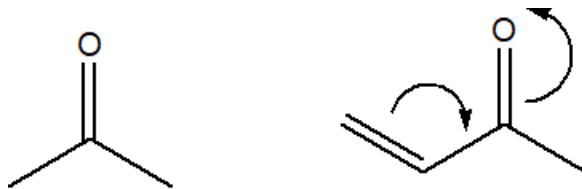
Number of molecules contains same functional group & show similar peak above 1500 cm^{-1} but they show different peak in finger print region. Therefore, we can say that each and every molecule have unique peak or band which is observed in fingerprint region, it is just like the finger print of human.

IR correlation chart

Types of vibration		Frequency (cm ⁻¹)
Alkane	C-C stretching	1200
	C-H stretching	3000-2840
	-CH ₂ bending	1465
	-CH ₃ bending	1375
	CH ₂ rocking	720
Alkenes	C=C stretching	1650
	=C-H stretching	3095-3010
	=C-H bending	1000-650
Alkynes	C≡C stretching	2100
	≡C-H stretching	3300
	≡C-H bending	700-600
Aromatic	C=C stretching	1600, 1500, 1450
	=C-H stretching	3040-3010
	=C-H bending	Below 900
C=O stretching	Amide	1680
	Carboxylic acid	1710
	Ketone	1715
	Aldehyde	1725
	Ester	1735
	Acid chloride	1800
	Anhydride	1760 (I), 1810 (II)
C-O	Stretching	1300-1000
O-H	Alcohol, phenol	
	Free	3600
	H-bonded	3400-3200
	Carboxylic acid	3400-2400
Amine	-N-H stretching	3440 (as), 3350 (s)
	-N-H bending	1650-1580
	C-N stretching	1350-1000
Amide	-N-H stretching	3370 (as), 3150 (s)
	-N-H bending	1650-1560
Nitriles	C≡N	2250
Imines	C=N	1690-1640
Nitro group (NO ₂)	N=O	1550 (as), 1350 (s)
Alkyl halide (C-X)	C-F	1350
	C-Cl	750
	C-Br, C-I	<660

Factor affecting vibrational frequency

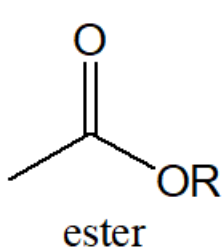
- a) Conjugation: As the conjugation increase, stretching frequency decreases, because force constant decreases due to conjugation.



$$\text{C}=\text{O (s)} = 1720 \text{ cm}^{-1} \quad \text{C}=\text{O (s)} = 1680 \text{ cm}^{-1}$$

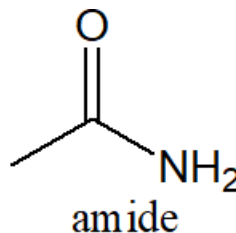
- b) Inductive effect and resonance effect:

Oxygen is more electronegative than nitrogen, therefore nitrogen easily donate electron or lone pair of nitrogen undergoes delocalization with C=O bond. Due to delocalization double bond of C=O changes into partial double bond therefore force constant decreases which decrease the C=O stretching frequency.



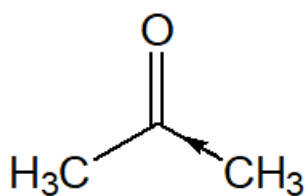
$$\text{C}=\text{O (s)} = 1735 \text{ cm}^{-1}$$

-I effect of -OR



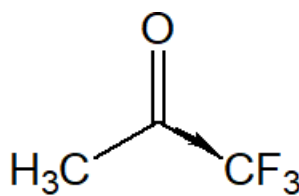
$$\text{C}=\text{O (s)} = 1690 \text{ cm}^{-1}$$

+R effect of -NH₂



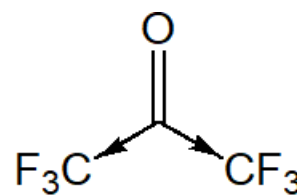
$$1720 \text{ cm}^{-1}$$

+I effect



$$1760 \text{ cm}^{-1}$$

-I effect



$$1790 \text{ cm}^{-1}$$

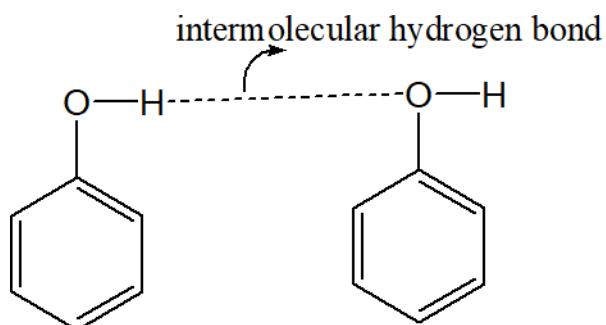
-I effect

- c) Hydrogen bonding:

Intermolecular hydrogen bonding weakens the O-H bond, thereby shifting the band to lower frequency. For example, in neat solution O-H stretching

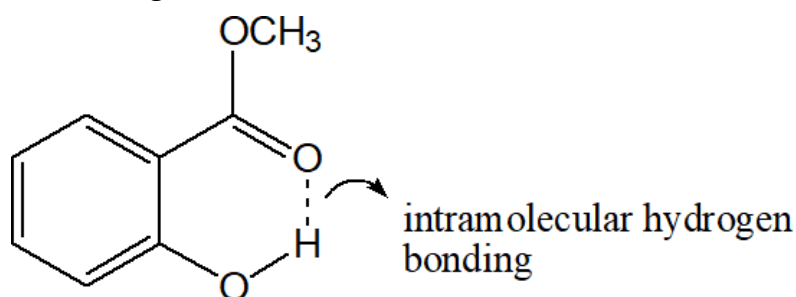
vibration of phenol observed in the range from $3400\text{--}3300\text{ cm}^{-1}$. When solution is dilute then O-H frequency shifted towards higher frequency at 3600 cm^{-1} .

Whereas in case of methyl salicylate, intramolecular hydrogen bonding lower down the



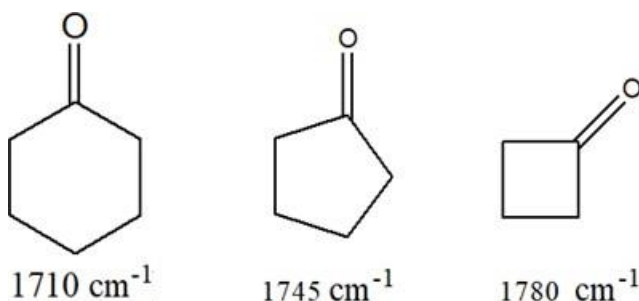
O-H frequency = $3400\text{--}3300\text{ cm}^{-1}$

stretching frequency of O-H at 3200 cm^{-1} . Intramolecular hydrogen bonding does not change its frequency even in very dilute solution because upon dilution structure of compound does not change.



O-H frequency centered at 3200 cm^{-1}

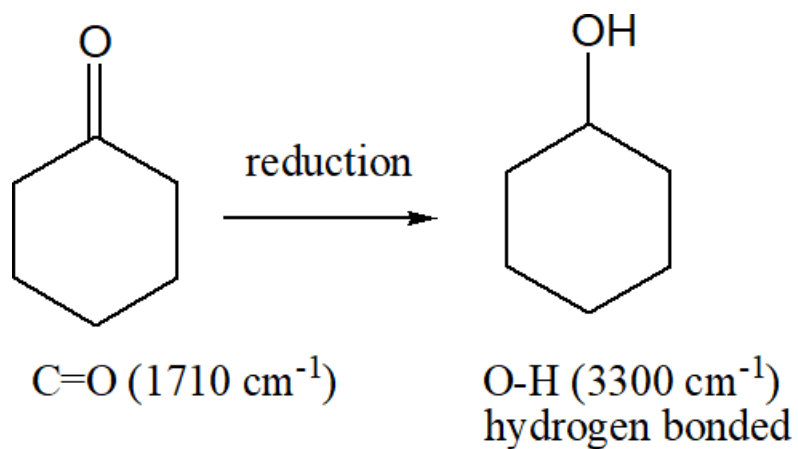
- d) Ring strain: As the size of the ring decrease, vibrational frequency of C=O increase.
For example.



Application of IR Spectroscopy

- a) Identification of different functional group.

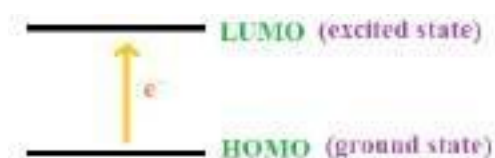
- b) Distinction between intermolecular and intra-molecular hydrogen bonding.
- c) Identification of purity of the compound, if compound is impure then additional peaks are observed in the IR spectrum.
- d) Study of chemical reaction:



- e) Identification of geometrical isomers (*cis-trans*).

UV-VISIBLE SPECTROSCOPY

- Ultraviolet and visible spectroscopy also known as electronic spectroscopy is used to measure the number of conjugations of double bond and aromatic conjugation within the molecule.
- It involves the promotion of electrons from HOMO to LUMO (HOMO means Highest Occupied Molecular Orbital whereas LUMO means Lowest Unoccupied Molecular Orbital).
- The HOMO-LUMO gap decrease as the conjugation increase.



- The ultraviolet region corresponds to 400-200 nm and visible region to 800-400 nm.
- UV-visible spectroscopy based on **Beer-Lambert law**.

$$\log \frac{I_0}{I} = \epsilon \cdot l \cdot c \text{ or } \epsilon = A / c l$$

Where,

I_0 = intensity of incident light

I = intensity of transmitted light

ϵ = molar absorptivity

l = path length of sample

c = concentration of sample

A = Absorbance

- Absorbance (A): It is reciprocal of Transmittance

$$\text{Absorbance (A)} = \text{Optical density (D)} = \log \frac{1}{T} \text{ or } \log \frac{I_0}{I}$$

- Transmittance (T): The fraction of incident light transmitted is known as Transmittance.

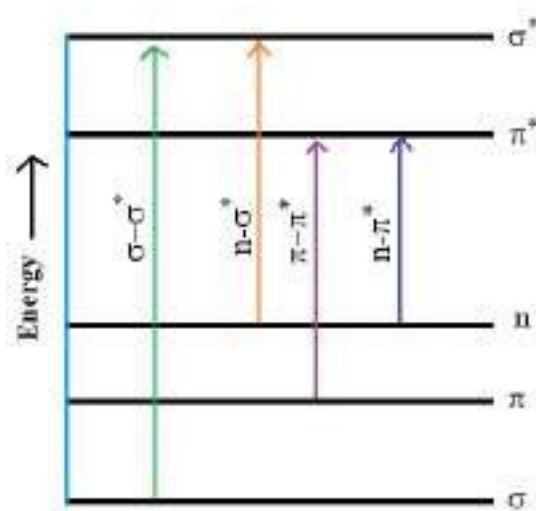
$$\text{Transmittance (T)} = \frac{I}{I_0}$$

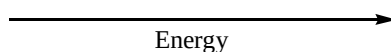
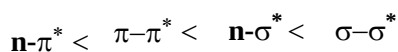
Electronic transitions

When molecule is getting excited by the absorption of electromagnetic radiation in UV-visible region then its electrons are promoted from ground state to excited state or from bonding orbital to anti-bonding orbital.

Types of electronic transitions

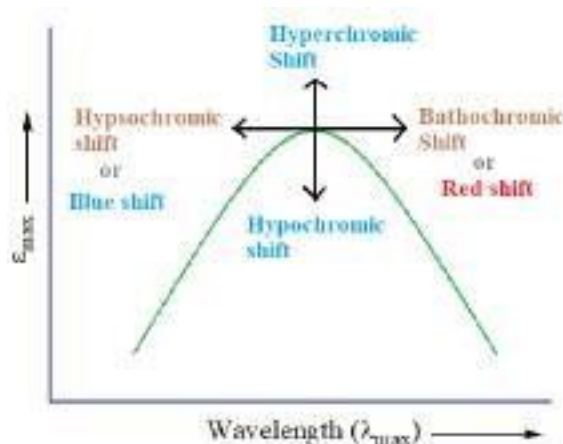
- $\sigma - \sigma^*$ Transition: transition of an electron from bonding sigma orbital (σ) to anti-bonding sigma orbital (σ^*), is represented by $\sigma - \sigma^*$ transition. For example, alkanes because in alkane all the atoms are held together with sigma bond.
- $n - \sigma^*$ & $n - \pi^*$ Transition: Transitions from non-bonding molecular (n) orbital to anti-bonding sigma orbital or anti-bonding pi orbital (π^*), are represented by $n - \sigma^*$ or $n - \pi^*$ transition respectively. This transition required less energy than $\sigma - \sigma^*$ transition. For example, alkyl halide, aldehydes, ketones etc.
- $\pi - \pi^*$ Transition: This type of transition generally shows in unsaturated molecules like alkenes, alkynes, aromatics, carbonyl compounds etc. This transition required less energy as compare to $n - \sigma^*$ transition.



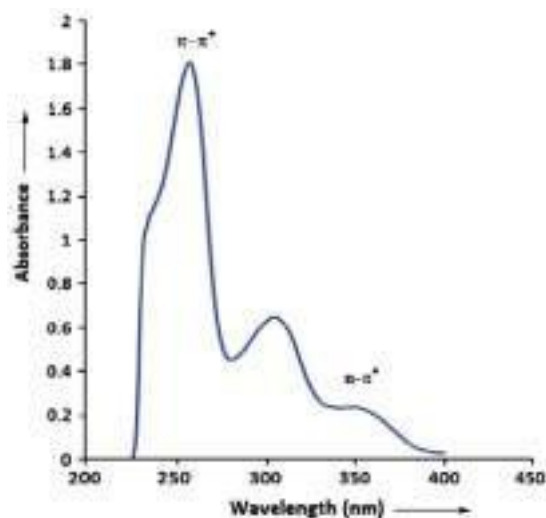


Absorption, intensity shift & UV spectrum

- Bathochromic shift: It is also known as Red shift, in this case absorption shift towards longer wavelength ($\lambda_{\max}$).
- Hypsochromic shift: It is also known as Blue shift, in this case absorption shift towards shorter wavelength ($\lambda_{\max}$).
- Hyperchromic shift: Intensity of absorption maximum ($\epsilon_{\max}$) increases.
- Hypochromic shift: Intensity of absorption maximum ($\epsilon_{\max}$) decrease.



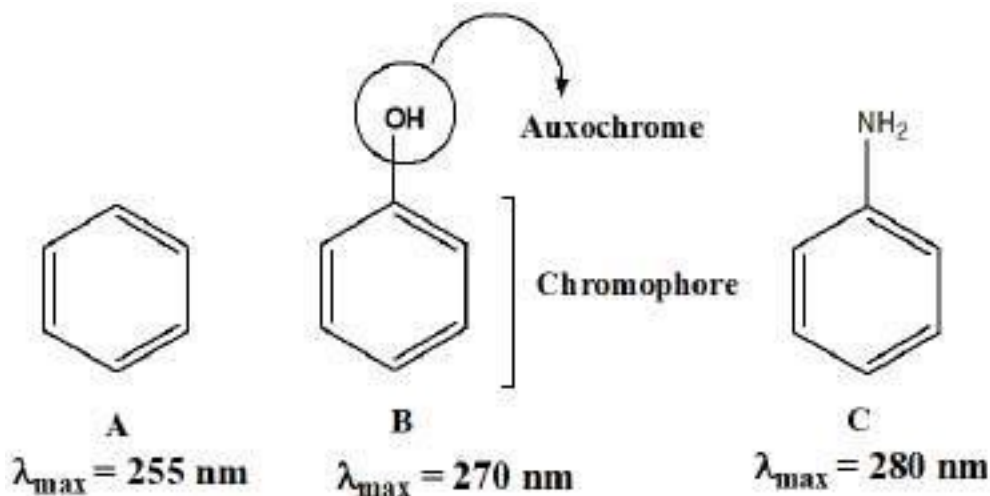
Absorption & intensity shift



UV spectrum of unknown compound

Chromophores & Auxochromes

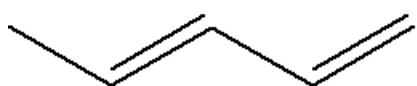
- a) Chromophores: Chromophore is a covalently unsaturated group absorbed electromagnetic radiation in UV-visible region and impart color to the compound. For example, $C=C$, $C\equiv C$, benzene, NO_2 etc.
- b) Auxochromes: Auxochrome is a saturated group (containing lone pair of electron), when auxochrome attach to the Chromophore absorption shift towards longer wavelength. For example, $-OH$, $-SH$, $-NH$ etc.



Factor affecting absorbance & intensity

a) Conjugation

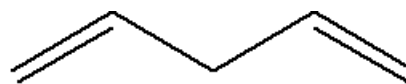
Absorption shift towards longer wavelength, if double bonds (chromophore) present in the molecule are in conjugation. For example,



$$\lambda_{\max} = 217 \text{ nm}$$

conjugated

A



$$\lambda_{\max} = 175 \text{ nm}$$

non-conjugation

B

In compound 'A', double bonds are in conjugation therefore 'A' possessing higher wavelength ($\lambda_{\max}$) as compare to compound 'B' (non-conjugated derivative).

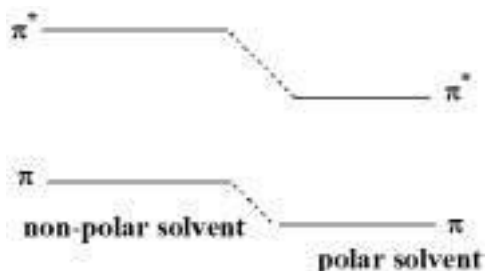
We know that,

$$E = \frac{hc}{\lambda}$$

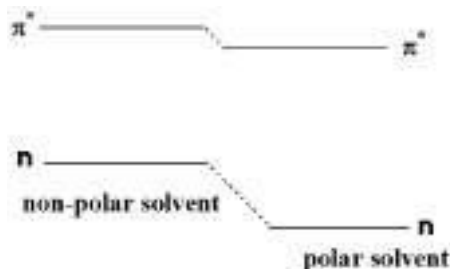
As the conjugation increase transition energy (E) between the orbitals is decrease and therefore wavelength ($\lambda_{\max}$) increase.

b) Effect of solvent:

In polar solvent, $\pi - \pi^*$ transitions shift towards longer wavelength (Red shift), because dipole interactions with polar solvent molecules lower the energy of excited state (π^*) than that of the ground state (π^* orbital's are stabilized by hydrogen bonding with polar solvent).



Whereas in polar solvent, $n - \pi^*$ transitions shift towards lower wavelength (Blue shift), because dipole interactions with polar solvent molecules lower the energy of ground state (n) than that of the excited state (n orbital's are stabilized by hydrogen bonding with polar solvent).

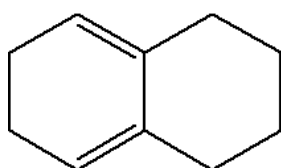


The Woodward-Fieser rules (Calculation of $\lambda_{\max}$ (nm) in conjugated dienes)

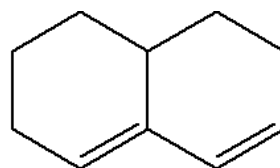
	$\lambda_{\max}$ (nm)
Base value:	
Acyclic or heteroannular dienes	214
Homoannular dienes	253
Increments for:	

Double bond extending conjugation	30
R alkyl substituent or ring residue	5
Exocyclic double bond	5
Polar groupings:	
-OCOCH ₃	0
-OR	6
-Cl, -Br	5
-NR ₂	60

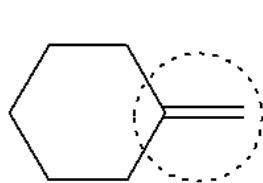
Explanation of terminology used in Woodward-Fieser rules



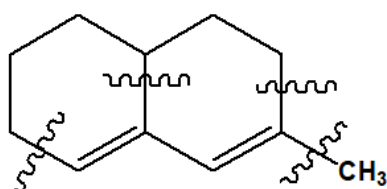
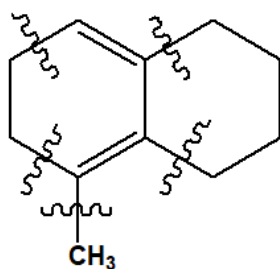
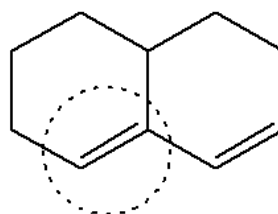
homoannular diene



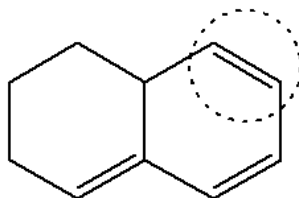
heteroannular diene



double bond exocyclic to the double bond

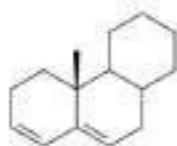


~~~~~ sign represent R alkyl group or ring residue

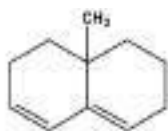


**Double bond extending conjugation**

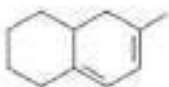




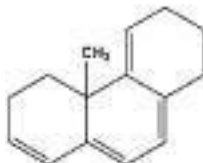
|                        |        |
|------------------------|--------|
| Base value:            | 214 nm |
| alkyl groups: 3x5      | 15 nm  |
| Exocyclic double bond: | 5 nm   |
| calculated:            | 234 nm |



|                        |        |
|------------------------|--------|
| Base value:            | 214 nm |
| Ring residue: 3 x 5    | 15 nm  |
| Exocyclic double bond: | 5 nm   |
| Calculated:            | 234 nm |



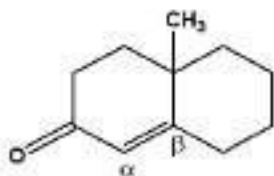
|                        |        |
|------------------------|--------|
| Base value:            | 255 nm |
| alkyl groups: 4x5      | 20 nm  |
| Exocyclic double bond: | 5 nm   |
| calculated:            | 278 nm |



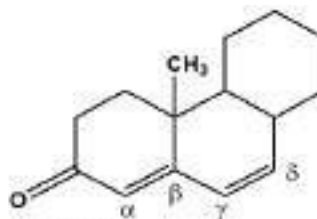
|                                           |        |
|-------------------------------------------|--------|
| Base value:                               | 255 nm |
| Ring residue: 5 x 5                       | 25 nm  |
| Exocyclic double bond: 3 x 5              | 15 nm  |
| double bond extending conjugation: 2 x 30 | 60 nm  |
| Calculated:                               | 353 nm |

### The Woodward-Fieser rules (Calculation of $\lambda_{\max}$ (nm) in $\alpha,\beta$ -unsaturated carbonyl compounds)

|                                     | $\lambda_{\max}$ (nm)                                    |
|-------------------------------------|----------------------------------------------------------|
| <b>Base value:</b>                  |                                                          |
| Acyclic or 6-membered cyclic ketone | 215                                                      |
| 5-membered cyclic ketone            | 202                                                      |
| Aldehyde                            | 210                                                      |
| <b>Increments for:</b>              |                                                          |
| Homocyclic diene component          | 39                                                       |
| Double bond extending conjugation   | 30                                                       |
| R alkyl substituent or ring residue | $\alpha = 10, \beta = 12, \gamma \text{ \& higher} = 18$ |
| Exocyclic double bond               | 5                                                        |
| <b>Polar groupings:</b>             |                                                          |
| -OH                                 | $\alpha = 35, \beta = 30, \gamma = 30$                   |
| -OR                                 | $\alpha = 35, \beta = 30, \gamma = 17$                   |
| -Cl,                                | $\alpha = 15, \beta = 12, \gamma = 12$                   |
| -Br                                 | $\alpha = 25, \beta = 30, \gamma = 25$                   |
| -NR <sub>2</sub>                    | $\beta = 95$                                             |
| -OCOCH <sub>3</sub>                 | $\alpha = 6, \beta = 6, \gamma = 6$                      |



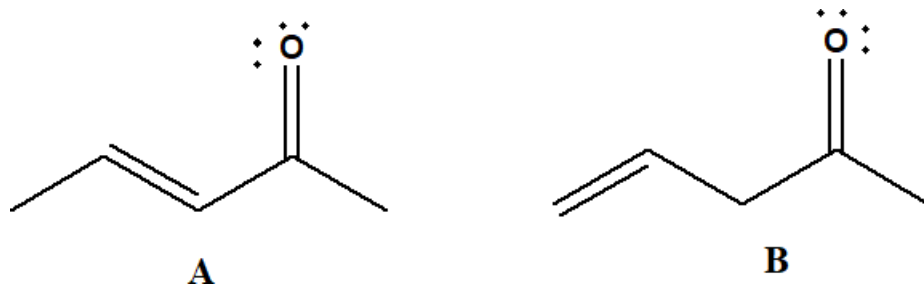
|                              |        |
|------------------------------|--------|
| Base value:                  | 215 nm |
| $\beta$ Ring residue: 2 x 12 | 24 nm  |
| Exocyclic double bond:       | 5 nm   |
| Calculated:                  | 244 nm |



|                                           |        |
|-------------------------------------------|--------|
| Base value:                               | 215 nm |
| $\beta$ Ring residue:                     | 12 nm  |
| $\delta$ Ring residue:                    | 18 nm  |
| Exocyclic double bond: 1 x 5              | 5 nm   |
| double bond extending conjugation: 1 x 30 | 30 nm  |
| Calculated:                               | 280 nm |

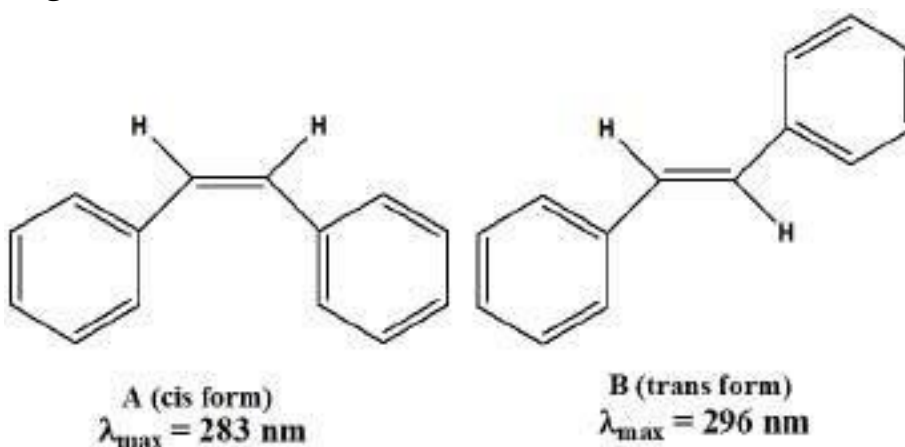
## Application of UV spectroscopy

- i) Extent of conjugation: As the double bond increase, absorption shift towards longer wavelength.
- ii) Distinction between conjugated and non conjugated compounds:



Both the above compound A & B are isomer of each other. Due to  $n - \pi^*$  transition, compound A appear at longer wavelength than compound B.

- iii) Study of geometric isomerism:



In isomer A, due to steric hindrance, absorption shift towards lower wavelength. We know that, steric hindrance destabilizes the compound means energy of the molecule increase which decreases the wavelength.

- iv) Used in the detection of impurities: Due to presence of impurities in the compound, give additional peaks are observed in UV spectrum.

### **INSTRUMENTATION:**

**Source:** Supplied light in the UV range (200-400 nm) hydrogen or deuterium lamp and Visible range (400-800 nm) Tungsten lamp.

**Monochromator:** Allows only the chosen wave length and absorb all the other wave length and this happen for the complete chosen range of wave length. Eg Prism, diffraction etc.

**Beam splitter:** Splits the beam of light of each wavelength into two halves of equal intensities. One goes to the test sample and other to the blank /reference sample.

**Cuvettes:** One cell is filled with the test sample and the other with the reference sample.

**Detector:** It detects the intensities of the transmittance light coming out of the cells and compares for each wave length and generates currents proportional to the differences in the intensities of the sample and blank.

**Recorder:** It records an Uv/ visible spectrograph with absorbance against the wave length.

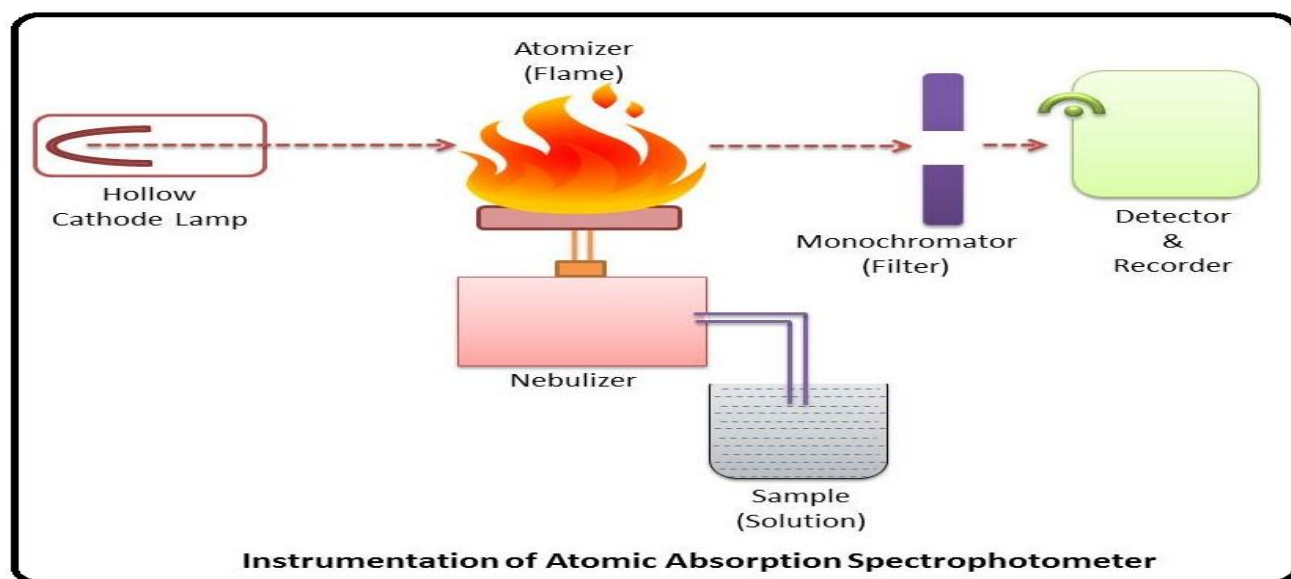
## ATOMIC ABSORPTION SPECTROSCOPY

- Atomic absorption spectroscopy is a very common technique for detecting metals and metalloids in samples.
- It is very reliable and simple to use.
- It also measures the concentration of metals in samples.

**Principal:** The basic principles of AAS can be expressed as follows. Firstly, all atoms or ions can absorb light at specific, unique wavelengths. When a sample containing copper (Cu) and nickel (Ni), for example, is exposed to light at the characteristic wavelength of Cu, then only the Cu atoms or ions will absorb this light. The amount of light absorbed at this wavelength is directly proportional to the concentration of the absorbing ions or atoms.

The electrons within an atom exist at various energy levels. When the atom is exposed to its own unique wavelength, it can absorb the energy (photons) and electrons move from a ground state to excited states. The radiant energy absorbed by the electrons is directly related to the transition that occurs during this process. Furthermore, since the electronic structure of every element is unique, the radiation absorbed represents a unique property of each individual element and it can be measured.

**Instrumentation:** An atomic absorption spectrometer uses these basic principles and applies them in practical quantitative analysis. A typical atomic absorption spectrometer consists of four main components: the light source, the atomization system, the monochromator and the detection system.



Namrata Heda





Uv